

Introduction to Contract Theory and Auctions
Lecture 6
Screening: General Models

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- A contracting problem with **Hidden Information**
- Uninformed party (principal) offers contract to informed party (agent)
- Examples
 - Insurance
 - Insuree knows her risk, insurer does not
 - Insurer offers several packages with different premiums and deductibles
 - Finance
 - Borrower knows the risk of project, lender does not
 - Lender offers several packages with different interest rates and collateral requirements
 - Hiring
 - Applicants know their ability, employer does not
 - Employer offers different packages varying wages, bonuses, etc.
 - Pricing
 - Buyer knows her valuation of the product, seller does not
 - Seller offers different qualities at different prices, or quantity discounts
- First analyzed formally, within the context of optimal income taxation, by Mirrlees, J. (1971).

- A monopolist is uninformed about the buyers' characteristics.
- What is the best pricing schedule according to quantity, quality, etc.?
- References: Mussa and Rosen (1978) & Maskin and Riley (1984).
- Although we will use monopoly as an example, similar results hold in other screening models.
- We will still use a relatively simple model but we will be more rigorous than before in our analysis.

Two-type model

- Monopolist produces quantity (or quality) $q \geq 0$ at cost $c(q)$
- Cost function is C^2 (twice-continuously differentiable) with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$
- If monopolist sells a quantity q for t dollars, his payoff is: $w(q, t) = t - c(q)$
- Customer has two possible taste parameters $\theta_H > \theta_L > 0$.
 - Probability of θ_L is $p \in (0, 1)$.
- If Type θ_i buys quantity q and pays t , her payoff is $u(q, t, \theta_i) = v(q, \theta_i) - t$
- The valuation function v is twice C^2 with respect to q with

$$\begin{aligned}v(0, \theta_i) &= 0, v_1 > 0, v_{11} \leq 0 \\v(q, \theta_H) &> v(q, \theta_L) \quad \text{for any } q > 0\end{aligned}$$

Notation

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued k -differentiable function we write

$$\frac{\partial f}{\partial x_i}(x) = f_i(x), \quad \frac{\partial^2 f}{\partial x_j \partial x_i}(x) = f_{ij}(x), \quad \frac{\partial^3 f}{\partial x_\ell \partial x_j \partial x_i}(x) = f_{ij\ell}(x)$$

and so on.

Two-type Model: Surplus

- Total surplus available is $S(q, \theta_i) = v(q, \theta_i) - c(q)$
- Assume that trade is desirable

$$S_1(0, \theta_i) > 0 \text{ for } i = L, H.$$

- But there can be too much of a good thing, i.e. $\exists \bar{q}$ such that

$$S(q, \theta_i) < 0 \forall q \geq \bar{q}$$

- Customer does not have to pay if she does not buy.
- The monopolist chooses how much to produce and possibly a non-linear price schedule.

Monopolistic Screening: First Best

- This is the case when **the monopolist knows the type of the consumer.**
- The monopolist solves the following problem for each $i = L, H$

$$\begin{aligned} \max_{q_i, t_i} \quad & t_i - c(q_i) \\ \text{s.t.} \quad & v(q_i, \theta_i) - t_i \geq 0. \end{aligned}$$

- The constraint must hold with equality, thus the problem becomes:

$$\max_{q_i \geq 0} S(q_i, \theta_i)$$

- The too much good assumption comes into play here: since $S(0, \theta_i) = 0 > S(q, \theta_i)$ for all $q \geq \bar{q}$ the previous problem is equivalent to

$$\max_{q_i} S(q_i, \theta_i) \quad \text{subject to } 0 \leq q_i \leq \bar{q}$$

- Since the objective function is continuous and the constraint set is compact, then, by Weierstrass's Theorem, a solution to this problem exists.

Theorem 6.1 (Weierstrass's Theorem)

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be compact and $f: \mathcal{D} \rightarrow \mathbb{R}$ be a continuous function on $\mathcal{D}$. Then f attains a maximum and a minimum on $\mathcal{D}$

Theorem 6.2 (Heine-Borel)

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **compact** if and only if it is closed and bounded.

Definition 6.3

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **bounded** if there exists $k > 0$ such that $\|x\| \leq k$ for every $x \in \mathcal{D}$.

Here $\|x\|$ represents any norm in $\mathbb{R}^n$. Take for example $\|x\| = (\sum_{i=1}^n x_i^\rho)^{1/\rho}$ for $\rho \in [1, \infty)$. if $\rho = 2$ then we are talking about the usual **Euclidean norm**.

Definition 6.4

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **closed** if and only if for all sequences (x^k) such that $x^k \in \mathcal{D}$ for each k and $x^k \rightarrow x$ then $x \in \mathcal{D}$

Definition 6.5

Given an norm $\|\cdot\|$ on $\mathcal{D} \subseteq \mathbb{R}^n$ define the **distance induced by the norm** between $x, y \in \mathbb{R}^n$ as $d(x, y) = \|x - y\|$

Definition 6.6

A sequence x^k in $\mathcal{D} \subseteq \mathbb{R}^n$ is said to converge to a limit x (written $x^k \rightarrow x$) if for all $\epsilon > 0$ there exists an integer $K(\epsilon)$ such that for all $k \geq K(\epsilon)$ we have $d(x^k, x) < \epsilon$.

Monopolistic Screening: First-Best Solution

- Let q_i^f be a solution
 - $q_i = 0$ cannot be a solution because trade at a positive quantity generates surplus, i.e. $S_1(0, \theta_i) > 0$.
 - Similarly, $q_i = \bar{q}$ cannot be a solution since surplus is negative at $\bar{q}$.
 - Thus, the solution must be interior:

$$S_1(q_i^f, \theta_i) = 0 \Leftrightarrow v_1(q_i^f, \theta_i) = c'(q_i^f), \quad i \in \{L, H\}$$

- If v is strictly concave on q ($v_{11} < 0$) or c is strictly convex ($c'' > 0$) then the solution is unique.
- We can solve for the transfers from the constraint as

$$t_i^f = v(q_i^f, \theta_i) \quad i \in \{L, H\}$$

- Features:
 - Monopolists maximizes surplus and captures it all.
 - Efficient quantities are produced.
 - Price is set to capture entire consumer surplus.
 - This is **first-degree price discrimination**.

- We can say more about the solution to the problem if we assume that.

$$v_1(q, \theta_H) > v_1(q, \theta_L)$$

- This condition is called the **single crossing** or **sorting** condition.
- The slope of the indifference curve for the high type is greater than the one for the low type.
 - Note that an indifference curve is $\bar{u} = v(q, \theta_i) - t$ for some $\bar{u}$.
 - Therefore, at any $(\tilde{q}, \tilde{t})$, $\frac{dt}{dq} = v_1(\tilde{q}, \tilde{t})$.
 - An equivalent condition is **Increasing Differences**: If $q > q'$ and $\theta > \theta'$ then

$$v(q, \theta) - v(q', \theta) > v(q, \theta') - v(q', \theta')$$

- Thus in this environment: Sorting condition $\Leftrightarrow$ Increasing differences.
- Easy to prove: For $(\Rightarrow)$ use the Fundamental Theorem of Calculus; for $(\Leftarrow)$ use the definition of a derivative.

Single-Crossing Condition

- $v(q, \theta_H) \geq v(q, \theta_L)$ implies that for any q , type H 's indifference curve is above type L 's.
- $v_{11}(q, \theta_i) \leq 0$ implies that the indifference curves are concave.
- Single-crossing condition implies that
 - at any point, type- H indifference curve has a higher slope than type L .
 - two indifference curves intersect at most once.
- In more general models, the single-crossing condition is called the **Spence Mirrlees condition** and it takes the following form:

$$\frac{\partial}{\partial \theta} \left(\frac{u_q}{u_t} \right) > 0 (< 0) \quad \text{for any } (t, q, \theta)$$

Proposition 6.7 (Monotonicity of the first-best solution)

If single-crossing condition holds, then $q_H^f \geq q_L^f$ and $t_H^f \geq t_L^f$.

Proof.

Towards a contradiction, suppose that $q_H^f < q_L^f$. Then

$$c'(q_H^f) = v_1(q_H^f, \theta_H) > v_1(q_H^f, \theta_L) \geq v_1(q_L^f, \theta_L) = c'(q_L^f)$$

where the strict inequality is from single-crossing and the weak inequality is by concavity of v . The previous chain of inequalities implies that $c'(q_H^f) > c'(q_L^f)$ but since c is convex, this implies that $q_H^f \geq q_L^f$ which is a contradiction.

Furthermore, since v is increasing and by single-crossing: $t_H^f = v(q_H^f, \theta_H) \geq v(q_L^f, \theta_L) = t_L^f$. $\square$

- Monopolist has to offer a common price schedule to both types and let them select.
- What happens if he offers the first best solution?
 - q_H^f at price t_H^f
 - q_L^f at price t_L^f
- Since $q_L^f > 0$, High type prefers (q_L^f, t_L^f)

$$v(q_L^f, \theta_H) - t_L^f = v(q_L^f, \theta_H) - v(q_L^f, \theta_L) > 0 = v(q_H^f, \theta_H) - t_H^f$$

- Everybody will buy low quantity and profit will be $t_L^f - c(q_L^f)$, which is less than the first best profit (Verify)
- What is the best the monopolist can do in this case?

- Thus the monopolist's problem now is to choose a price schedule $t(q)$ to maximize expected profits

$$\begin{aligned} \max_{t(q)} \quad & p(t(q_L) - c(q_L)) + (1 - p)(t(q_H) - c(q_H)) \\ \text{subject to} \quad & q_i \in \arg \max_{q \geq 0} \{v(q, \theta_i) - t(q)\} \quad \text{for } i = L, H \\ & v(q_i, \theta_i) - t(q_i) \geq 0 \quad \text{for } i = L, H \end{aligned}$$

- The problem is difficult as the choice set is a set of functions.
- **Revelation Principle** simplifies the problem tremendously
 - We can restrict the choice set to **menus** $\{(q_L, t_L), (q_H, t_H)\}$.

- In general, the monopolist need not constraint himself to nonlinear price schedules $t(q)$.
- he can set up a **mechanism** in which the customer chooses a strategy s from some set S and the monopolist assigns the outcome $g(q(s), t(s))$.
- A mechanism is a pair (S, g) , where $g : S \rightarrow D$ and $D = \{(q, t) : q \in \mathbb{R}_+, t \in \mathbb{R}\}$ is the set of outcomes.
- A mechanism implements an outcome $f : \Theta \rightarrow D$ if there exists $s^* : \Theta \rightarrow S$ such that for all $\theta \in \Theta$, $g(s^*(\theta)) = f(\theta)$ and

$$u(g(s^*(\theta)), \theta) \geq u(g(s'), \theta), \text{ for all } s' \in S$$

- A **direct (revelation) mechanism** is (Θ, g) ; i.e., the customer (agent) reports a type $\theta \in \Theta$ and the monopolist (principal) assigns outcome $g(\theta)$.
- A direct mechanism is **incentive compatible (truthful)** if reporting the true type is optimal, i.e., for all $\theta \in \Theta$

$$u(g(\theta), \theta) \geq u(g(\theta'), \theta), \text{ for all } \theta' \in \Theta$$

Theorem 6.8

If there exist a mechanism that implements an outcome, then there exists a truthful direct mechanism that implements the same outcome.

Proof.

Suppose the mechanism (S, g) implements f . Then there exists $s^* : \Theta \rightarrow S$ such that for all $\theta \in \Theta$, $g(s^*(\theta)) = f(\theta)$ and

$$u(g(s^*(\theta)), \theta) \geq u(g(s'), \theta), \text{ for all } s' \in S$$

which implies that

$$u(g(s^*(\theta)), \theta) \geq u(g(s^*(\theta')), \theta), \text{ for all } \theta' \in \Theta$$

Let the direct mechanism (Θ, f) . Then

$$u(f(\theta), \theta) = u(g(s^*(\theta)), \theta) \geq u(g(s^*(\theta')), \theta), \text{ for all } \theta' \in \Theta$$

and hence

$$u(f(\theta), \theta) \geq u(f(\theta'), \theta) \text{ for all } \theta' \in \Theta$$

Thus (Θ, f) is a truthful direct mechanism that implements the same outcome. □

Monopolistic Screening—Second Best

The revelation principle implies that the monopolist's problem can be formulated as

$$(P) \quad \max_{(q_L, t_L), (q_H, t_H)} p(t_L - c(q_L)) + (1 - p)(t_H - c(q_H)) \quad (1)$$

subject to

$$v(q_L, \theta_L) - t_L \geq v(q_H, \theta_L) - t_H \quad (2)$$

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \quad (3)$$

$$v(q_L, \theta_L) - t_L \geq 0 \quad (4)$$

$$v(q_H, \theta_H) - t_H \geq 0 \quad (5)$$

$$q_H \geq 0, q_L \geq 0 \quad (6)$$

- Constraints (2) and (3) are called **Incentive Compatibility** (IC) constraints.
- Constraints (4) and (5) are called **Individual Rationality** (IR) constraints.

Relaxed Program

- This problem is equivalent to a simpler one in which constraints are
 - IC constraint of type H binds.
 - IR constraint of type L binds.
 - $q_H \geq q_L$.
- We have seen this in the two simple examples we analyzed before.

$$(RP) \quad \max_{(q_L, t_L), (q_H, t_H)} p(t_L - c(q_L)) + (1 - p)(t_H - c(q_H)) \quad (7)$$

subject to

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \quad (8)$$

$$v(q_L, \theta_L) - t_L \geq 0 \quad (9)$$

$$q_H \geq q_L \quad (10)$$

$$q_H \geq 0, q_L \geq 0 \quad (11)$$

Proposition 6.9

Solution sets to the programs (P) and (RP) are the same.

- 1 The constraint set of (P) is a subset of that of (RP). (2) and (3) imply that

$$v(q_H, \theta_L) - v(q_L, \theta_L) \leq t_H - t_L \leq v(q_H, \theta_H) - v(q_L, \theta_H)$$

Single crossing condition implies that $q_H \geq q_L$.

- 2 At the solution to (RP), (8) holds as an equality: Otherwise we could increase t_H a little bit.
- 3 Solutions to (RP) satisfy the constraint of (P): since (8) holds with equality and $q_H \geq q_L$, single crossing implies

$$t_H - t_L = v(q_H, \theta_H) - v(q_L, \theta_H) \geq v(q_H, \theta_L) - v(q_L, \theta_L)$$

which implies (2). Also (8), (9) and $\theta_H \geq \theta_L$ imply (4):

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \geq v(q_L, \theta_L) - t_L \geq 0$$

- ① Let a^P solve (P) but not (RP)
 - Step (1) implies a^P satisfies the constraints of (RP).
 - At any solution a^R to (RP) the value of the objective function is strictly greater than its value at a^P .
 - Then, a^R is not in the constraint set of (P), contradicting step (3).
- ② Let a^R solve (RP) but not (P)
 - Step (3) implies a^R is in the constraint set of (P)
 - There is an a^P in the constraint set of (P) that gives strictly higher value for the objective function.
 - Step (1) implies a^P is in the constraint set of (RP)
 - Contradicts that a^R solves (RP)

This method is quite common:

- ① Find an easier “relaxed” problem by omitting difficult constraints.
- ② Verify that any solution to the relaxed problem satisfies the omitted constraints.

- Finally, note that (8) and (9) must hold with equality in the solution.
- Therefore, the problem is equivalent to:

$$\max_{(q_L, t_L), (q_H, t_H)} p(t_L - c(q_L)) + (1 - p)(t_H - c(q_H)) \quad (12)$$

subject to

$$v(q_H, \theta_H) - t_H = v(q_L, \theta_H) - t_L \quad (13)$$

$$v(q_L, \theta_L) - t_L = 0 \quad (14)$$

$$q_H \geq q_L \quad (15)$$

- Note that $t_L = v(q_L, \theta_L)$, i.e., the low type receives zero surplus and

$$t_H = v(q_H, \theta_H) - [v(q_L, \theta_H) - v(q_L, \theta_L)]$$

i.e., the high type obtains a surplus, known as **information rent**.

- Condition (15) is called **monotonicity**. The usual approach is to ignore it for the moment, and solve the new relaxed program.
- After we obtain the solution to the new relaxed program, we will verify that it satisfies (15).
- Substitute in the other constraints and the problem becomes

$$\max_{q_L \geq 0, q_H \geq 0} p[v(q_L, \theta_L) - c(q_L)] + (1 - p)[v(q_H, \theta_H) - c(q_H)] - (1 - p)[v(q_L, \theta_H) - v(q_L, \theta_L)]$$

- Compare this to the first best problem

$$\max_{q_L \geq 0, q_H \geq 0} p[v(q_L, \theta_L) - c(q_L)] + (1 - p)[v(q_H, \theta_H) - c(q_H)]$$

- First maximizes expected surplus.
- Second best maximizes expected surplus minus **expected information rent**.

- For reasons that will become clear later, it is convenient to rewrite the problem as:

$$\max_{q_L \geq 0, q_H \geq 0} v(q_L, \theta_L) - c(q_L) + \frac{1-p}{p} (v(q_H, \theta_H) - c(q_H)) - \frac{1-p}{p} (v(q_L, \theta_H) - v(q_L, \theta_L))$$

- Assume the following term (**called the virtual surplus**) is strictly concave.

$$S^v(q, \theta_L, \theta_H) = (v(q, \theta_L) - c(q)) - \frac{1-p}{p} (v(q, \theta_H) - v(q, \theta_L))$$

- This implies the objective is concave, so the first order conditions (Kuhn-Tucker) will be necessary and sufficient.

Form the Lagrangean:

$$\begin{aligned} L(q_L, q_H, \lambda_L, \lambda_H) = & v(q_L, \theta_L) - c(q_L) - \frac{1-p}{p}(v(q_L, \theta_H) - v(q_L, \theta_L)) \\ & + \frac{1-p}{p}(v(q_H, \theta_H) - c(q_H)) + \lambda_L q_L + \lambda_H q_H \end{aligned}$$

A critical point must satisfy the following conditions:

$$\begin{aligned} \frac{1-p}{p}(v_1(q_H, \theta_H) - c'(q_H)) + \lambda_H &= 0 \\ (v_1(q_L, \theta_L) - c'(q_L) - \frac{1-p}{p}(v_1(q_L, \theta_H) - v_1(q_L, \theta_L)) + \lambda_L &= 0 \\ \lambda_L, \lambda_H, q_L, q_H \geq 0, \quad \lambda_L q_L = \lambda_H q_H &= 0 \end{aligned}$$

Second Best solution

- $q_H = 0 \Rightarrow v_1(0, \theta_H) - c'(0) \leq 0$, contradicting $S_1(0, \theta_H) > 0$.
- Therefore, $q_H > 0 \Rightarrow \lambda_H = 0$ and hence

$$v_1(q_H, \theta_H) - c'(q_H) = 0$$

- Similarly, if $q_L > 0$

$$v_1(q_L, \theta_L) - c'(q_L) - \frac{1-p}{p}(v_1(q_L, \theta_H) - v_1(q_L, \theta_L)) = 0$$

- and if $q_L = 0$, $\lambda_L \geq 0$ and

$$v_1(0, \theta_L) - c'(0) - \frac{1-p}{p}(v_1(0, \theta_H) - v_1(0, \theta_L)) \leq 0$$

The unique global solution q_L^s, q_H^s is given by

$$v_1(q_H^s, \theta_H) = c'(q_H^s)$$

- If $S_1^v(0, \theta_L, \theta_H) \geq 0$

$$v_1(q_L^s, \theta_L) = c'(q_L^s) + \frac{1-p}{p}(v_1(q_L^s, \theta_H) - v_1(q_L^s, \theta_L))$$

- otherwise $q_L^s = 0$

The solution is also monotone

- We have to check if the solution satisfies constraint (15): $q_H^s \geq q_L^s$
- If $q_L^s = 0$ then it is satisfied trivially.
- So suppose $q_L^s > 0$. Then

$$v_1(q_L^s, \theta_L) = c'(q_L) + \frac{1-p}{p}(v_1(q_L^s, \theta_H) - v_1(q_L^s, \theta_L)) > c'(q_L^s)$$

- Remember $v_1(q_L^f, \theta_L) = c'(q_L^f)$. We claim that $q_L^s \leq q_L^f$. Towards a contradiction, $q_L^s > q_L^f$. Then, convexity of c implies that

$$v_1(q_L^s, \theta_L) > c'(q_L^s) \geq c'(q_L^f) = v_1(q_L^f, \theta_L)$$

- But concavity of v in q , i.e. $v_{11} \leq 0$, implies that $q_L^s \leq q_L^f$, a contradiction.
- Therefore we have $q_H^s = q_H^f \geq q_L^f \geq q_L^s$ and thus (15) is satisfied.

Second Best compared to First Best

Remember that the first best policy was given by $v_1(q_i^f, \theta_i) = c'(q_i^f)$.

- In the second best solution the high type consumes the efficient quantity but the low type's consumption is lower than the efficient level.
- The high type obtains an information rent.

The problem with the efficient allocation was that high type preferred the low type's allocation. What do you do to solve this problem?

- Keep giving the high type the same quantity at a lower price.
 - This gives her a positive surplus, i.e., information rent.
- Reduce quantity and price for the low type.
 - You do this to keep high type's information rent at minimum.
- Distortion in low type's consumption is decreasing in p .

- Suppose there are $n \geq 2$ types such that:

$$\theta_n > \theta_{n-1} > \cdots > \theta_1$$

- Assume that $\Pr(\theta_i) = p_i > 0$, $\sum_{i=1}^n p_i = 1$.
- The rest of the model is the same:

$$v(q, \theta_{i+1}) \geq v(q, \theta_i), v(0, \theta_i) = 0, v_1 > 0, v_{11} \leq 0, v_1(q, \theta_{i+1}) > v_1(q, \theta_i)$$

- Results generalize
 - No distortion at the top.
 - Inefficiently low consumption for lower types.
 - No rent for the lowest type. Higher types receive information rent.
 - Downward incentives compatibility constraints bind.

The monopolist's problem is:

$$\max_{\{(q_i, t_i)\}_{i=1, n}} \sum_{i=1}^n p_i(t_i - c(q_i))$$

subject to

$$v(q_i, \theta_i) - t_i \geq v(q_j, \theta_i) - t_j, \quad \forall i, j \quad (16)$$

$$v(q_i, \theta_i) - t_i \geq 0, \quad \forall i \quad (17)$$

$$q_i \geq 0, \quad \forall i \quad (18)$$

The general problem is equivalent to the following relaxed problem:

$$\max_{\{(q_i, t_i)\}_{i=1, n}} \sum_{i=1}^n p_i(t_i - c(q_i))$$

subject to

$$v(q_i, \theta_i) - t_i \geq v(q_{i-1}, \theta_i) - t_{i-1}, \quad \forall i = 2, \dots, n \quad (19)$$

$$v(q_1, \theta_1) - t_1 \geq 0 \quad (20)$$

$$q_i \geq q_{i-1}, \quad i = 2, \dots, n \quad (21)$$

$$q_i \geq 0, \quad \forall i \quad (22)$$

First, we show that when $q_i \geq q_{i-1}$ holds, then (16) holds if and only if (19) and

$$v(q_i, \theta_i) - t_i \geq v(q_{i+1}, \theta_i) - t_{i+1}, \quad i = 1, \dots, n-1. \quad (23)$$

Clearly, (16) imply (19) and (23). To prove the other direction, fix an index $i \in \{2, \dots, n\}$ and take any $j < i$. Note that

$$\begin{aligned} v(q_i, \theta_i) - v(q_j, \theta_i) &= \sum_{k=j+1}^i v(q_k, \theta_i) - v(q_{k-1}, \theta_i) \\ &\geq \sum_{k=j+1}^i v(q_k, \theta_k) - v(q_{k-1}, \theta_k) \\ &\geq \sum_{k=j+1}^i t_k - t_{k-1} = t_i - t_j \end{aligned}$$

Similarly, for any $j > i$ (verify this!)

- 1 The Constraint set of (P) is a subset of that of (RP): (16) implies that

$$v(q_i, \theta_{i-1}) - v(q_{i-1}, \theta_{i-1}) \leq t_i - t_{i-1} \leq v(q_i, \theta_i) - v(q_{i-1}, \theta_i)$$

Single crossing condition implies that $q_i \geq q_{i-1}$.

- 2 At the solution to (RP), (19) holds as an equality for all i : Otherwise, we can increase all t_j , $j \geq i$, by some $\epsilon > 0$.
- 3 Solutions to (RP) satisfy the constraints of (P): Since (19) hold with equality and $q_i \geq q_{i-1}$, single crossing implies

$$t_i - t_{i-1} = v(q_i, \theta_i) - v(q_{i-1}, \theta_i) \geq v(q_i, \theta_{i-1}) - v(q_{i-1}, \theta_{i-1})$$

which implies (23). Our result from the previous slide implies (16). Showing that (16) and (20) imply (17) is left for you to verify.

Clearly, constraints (19) and (20) are all binding. Therefore, the problem is

$$\begin{aligned} \max_{\{(q_i, t_i)\}_{i=1, n}} \quad & \sum_{i=1}^n p_i(t_i - c(q_i)) \\ \text{subject to} \quad & v(q_i, \theta_i) - t_i = v(q_{i-1}, \theta_i) - t_{i-1}, \quad \forall i = 2, \dots, n \\ & v(q_1, \theta_1) - t_1 = 0 \\ & q_i \geq q_{i-1}, \quad i = 2, \dots, n \\ & q_i \geq 0, \quad \forall i \end{aligned}$$

As before, ignore for now the monotonicity constraint and form the Lagrangian

$$\begin{aligned}\mathcal{L} = & \sum_{i=1}^n p_i(t_i - c(q_i)) + \sum_{i=2}^n \lambda_i[v(q_i, \theta_i) - v(q_{i-1}, \theta_i) - t_i + t_{i-1}] \\ & + \lambda_1[v(q_1, \theta_1) - t_1] + \sum_{i=1}^n \mu_i q_i\end{aligned}$$

The Kuhn-Tucker conditions for optimality are:

$$\frac{\partial \mathcal{L}}{\partial q_i} = 0 \quad i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial t_i} = 0 \quad i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_i} = 0 \quad i = 1, \dots, n$$

$$\mu_i q_i = 0, \mu_i \geq 0, q_i \geq 0 \quad i = 1, \dots, n$$

- We will focus on solutions such that $q_i > 0$ for all i .
- From the KT conditions, this implies that $\mu_i = 0$ for all i .

From the previous slide, the KT conditions imply that

- For $i < n$

$$\frac{\partial \mathcal{L}}{\partial q_i} = -p_i c'(q_i) + \lambda_i v_1(q_i, \theta_i) - \lambda_{i+1} v_1(q_i, \theta_{i+1}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial t_i} = p_i - \lambda_i + \lambda_{i+1} = 0$$

- For $i = n$

$$\frac{\partial \mathcal{L}}{\partial q_n} = -p_n c'(q_n) + \lambda_n v_1(q_n, \theta_n) = 0$$

$$\frac{\partial \mathcal{L}}{\partial t_n} = p_n - \lambda_n = 0$$

- So, we have no distortion at the top:

$$v_1(q_n, \theta_n) = c'(q_n)$$

- Using $p_i - \lambda_i + \lambda_{i+1} = 0$ and $\lambda_n = p_n$ we get $\lambda_i = \sum_{k=i}^n p_k$
- Therefore, for $i < n$

$$v_1(q_i, \theta_i) - c'(q_i) = \frac{1 - P_i}{p_i} [v_1(q_i, \theta_{i+1}) - v_1(q_i, \theta_i)] > 0$$

where $P_i = \sum_{k=1}^i p_k$

- Note that the principal maximizes the sum of virtual surpluses:

$$S^v(q_i, \theta_i, \theta_{i+1}) = v(q_i, \theta_i) - c(q_i) - \frac{1 - P_i}{p_i} [v(q_i, \theta_{i+1}) - v(q_i, \theta_i)]$$

- This implies less than efficient q_i for all $i < n$.

- If the virtual surplus functions are strictly concave and
 - $(1 - P_i)/p_i$ (inverse hazard rate) decreasing in i
 - Known as **monotone hazard rate property**
 - $v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i) \geq v_1(q_{i+1}, \theta_{i+2}) - v_1(q_{i+1}, \theta_{i+1})$ then $q_{i+1} \geq q_i$.
 - Proof: Towards a contradiction, suppose that $q_{i+1} < q_i$. Then we have that

$$\begin{aligned}
 0 &= v_1(q_i, \theta_i) - c'(q_i) - \frac{1 - P_i}{p_i} [v_1(q_i, \theta_{i+1}) - v_1(q_i, \theta_i)] \\
 &< v_1(q_{i+1}, \theta_i) - c'(q_{i+1}) - \frac{1 - P_i}{p_i} [v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i)] \\
 &\leq v_1(q_{i+1}, \theta_{i+1}) - c'(q_{i+1}) - \frac{1 - P_i}{p_i} [v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i)] \\
 &\leq v_1(q_{i+1}, \theta_{i+1}) - c'(q_{i+1}) - \frac{1 - P_{i+1}}{p_{i+1}} [v_1(q_{i+1}, \theta_{i+2}) - v_1(q_{i+1}, \theta_{i+1})] = 0
 \end{aligned}$$

which is a contradiction.

- However, it is possible to have $q_i = 0$ for some $i < n$.
- In most papers, set of types is a continuum, e.g. $[\theta_L, \theta_H]$
 - More technical but the results will be cleaner.

- Type set: $\Theta = [\theta_L, \theta_H]$
- Cumulative distribution function of $\theta : F$, continuous density $f(\theta) > 0$.
- $v_1, v_2, v_{12} > 0, v_{11} \leq 0$
- A contract is a pair of functions $(q, t) : \theta \rightarrow R_+ \times R$

Continuum of types

The monopolist problem is

$$\max_{(q,t)} \int_{\theta_L}^{\theta_H} [t(\theta) - c(q(\theta))] f(\theta) d\theta$$

subject to

$$v(q(\theta), \theta) - t(\theta) \geq v(q(\theta'), \theta) - t(\theta'), \forall \theta, \theta'$$

$$v(q(\theta), \theta) - t(\theta) \geq 0, \forall \theta$$

For any contract (q, t) let

$$U(\theta) = v(q(\theta), \theta) - t(\theta)$$

We first characterize incentive compatible contracts

Theorem 6.10

A contract (q, t) is incentive compatible if and only if

- ① *q is increasing*
- ② *$U(\theta) = U(\theta') + \int_{\theta'}^{\theta} v_2(q(\tau), \tau) d\tau$ for any θ, θ' .*

⇒ We can write IC constraints of types $\theta' > \theta$ as

$$\begin{aligned} U(\theta) &\geq U(\theta') + v(q(\theta'), \theta) - v(q(\theta'), \theta') \\ U(\theta') &\geq U(\theta) + v(q(\theta), \theta') - v(q(\theta), \theta) \end{aligned}$$

Therefore

$$v(q(\theta), \theta') - v(q(\theta), \theta) \leq U(\theta') - U(\theta) \leq v(q(\theta'), \theta') - v(q(\theta'), \theta)$$

- Single crossing then implies $q(\theta') \geq q(\theta)$ (verify)
- Monotonicity of q implies that it is continuous almost everywhere.

Also

$$\frac{v(q(\theta), \theta') - v(q(\theta), \theta)}{\theta' - \theta} \leq \frac{U(\theta') - U(\theta)}{\theta' - \theta} \leq \frac{v(q(\theta'), \theta') - v(q(\theta'), \theta)}{\theta' - \theta}$$

Taking limits as $\theta' \rightarrow \theta$

$$\frac{dU(\theta)}{d\theta} = v_2(q(\theta), \theta)$$

at all points where q is continuous. If U is absolutely continuous (a strong form of continuity), then

$$U(\theta) = U(\theta') + \int_{\theta'}^{\theta} v_2(q(\tau), \tau) d\tau \quad \forall \theta, \theta' \quad (24)$$

This **integral** form of characterizing incentive compatibility is actually a result from a more general type of results called **envelope theorems**. The particular version that applies here is quite powerful. Please see Milgrom and Segal (2002) Econometrica paper for the original result, Milgrom's textbook (in the syllabus), chapter 3, and my own brief notes on the topic (in Moodle).

⇐ Consider any $\theta' > \theta$.

$$\begin{aligned} U(\theta') - U(\theta) &= \int_{\theta}^{\theta'} v_2(q(\tau), \tau) d\tau \\ &\geq \int_{\theta}^{\theta'} v_2(q(\theta), \tau) d\tau \\ &= v(q(\theta), \theta') - v(q(\theta), \theta) \end{aligned}$$

where the inequality follows from SCP and q increasing.

Also note that

$$\begin{aligned} U(\theta) &= U(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \\ &\geq U(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\theta_L), \tau) d\tau \\ &= U(\theta_L) + v(q(\theta_L), \theta) - v(q(\theta_L), \theta_L) \\ &\geq 0 \end{aligned}$$

again by SCP, the fact that q is increasing, and $v_2 > 0$. Therefore, we need only to consider PC constraint for type θ_L .

Therefore, the problem is equivalent to

$$\max_{(q,t)} \int_{\theta_L}^{\theta_H} (t(\theta) - c(q(\theta))) f(\theta) d\theta \quad (25)$$

$$\text{subject to} \quad (26)$$

$$U(\theta) = U(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \quad \forall \theta \quad (27)$$

$$q \text{ is increasing} \quad (28)$$

$$U(\theta_L) \geq 0 \quad (29)$$

Using (27) we can write the objective function as

$$\int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \right] f(\theta) d\theta - U(\theta_L)$$

Integration by parts gives [Let $u = \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau$]

$$\int_{\theta_L}^{\theta_H} \left[\int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \right] f(\theta) d\theta = \int_{\theta_L}^{\theta_H} (1 - F(\theta)) v_2(q(\theta), \theta) d\theta$$

Therefore, the objective function can be written as

$$\int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_2(q(\theta), \theta) \right] f(\theta) d\theta - U(\theta_L)$$

Therefore, the monopolist's problem reduces to

$$\max_q \int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_2(q(\theta), \theta) \right] f(\theta) d(\theta)$$

subject to q increasing.

Under some regularity assumptions on the objective function (ask me if you are curious), we can differentiate under the integral sign to get the first order condition with respect to q for each θ :

$$v_1(q(\theta), \theta) - c'(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) = 0$$

Note that the monopolist maximizes virtual surplus!

- No distortion at the top:

$$v_1(q(\theta_H), \theta_H) = c'(q(\theta_H))$$

- Lower types $\theta < \theta_H$ under-consume since

$$\frac{1 - F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) > 0$$

- All types other than the lowest receive information rent.
- If virtual surplus function

$$v(q, \theta) - c(q) - \frac{1 - F(\theta)}{f(\theta)} v_2(q, \theta)$$

is strictly concave in q , then the solution is unique.

Monotonicity

If

- $\frac{1-F(\theta)}{f(\theta)}$ is decreasing (monotone hazard rate property)
- $\frac{\partial v_{12}}{\partial \theta} \leq 0$

then q is increasing. To see this, let $\theta' > \theta$ but $q(\theta') < q(\theta)$. Then

$$\begin{aligned} 0 &= v_1(q(\theta), \theta) - c'(q(\theta)) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) \\ &< v_1(q(\theta'), \theta) - c'(q(\theta')) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta'), \theta) \\ &\leq v_1(q(\theta'), \theta') - c'(q(\theta')) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta'), \theta) \\ &\leq v_1(q(\theta'), \theta') - c'(q(\theta')) - \frac{1-F(\theta')}{f(\theta')} v_{12}(q(\theta'), \theta') \\ &= 0 \end{aligned}$$

which is a contradiction.

Common results from the standard screening model:

- At the efficient allocation, high types want to mimic low types → first best is not incentive compatible.
- They have to receive (information) rents to prevent mimicking.
- At second best principal trades off efficiency for information rents.
 - Less than efficient for low types to reduce rent for high types.
 - Maximizes virtual surplus = surplus - information rent.
- Efficient consumption for the highest type.
- All types except the lowest receive information rent.
- Higher types consume more than lower ones.

- Multidimensional q and θ : we have seen a model with multidimensional θ . However, the case for multidimensional q and θ , except in special cases is extremely difficult to solve.
- Principal's payoff directly depends on θ : common values.
- Agent's reservation utility depends on θ .
- Multi-principal and/or multi-agent models (If we have time, I will talk about auctions).
- Dynamic models.